

Foundations of Artificial Intelligence

M. Helmert
C. Büchner

University of Basel
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Exercise Sheet 8 — Solutions

Exercise 8.1 (1+2 marks)

Consider the propositional formula $\varphi = ((A \vee \neg C) \rightarrow (\neg C \wedge (\neg A \vee B)))$.

- (a) Is φ satisfiable, unsatisfiable, falsifiable, valid? Briefly justify your answers.
- (b) Transform φ into CNF using the equivalences from Chapter E2. Provide all intermediate formulas that result from applying an equivalence transformation.

Solution:

- (a)
 - satisfiable: Yes, for example $\{A \mapsto \mathbf{F}, B \mapsto \mathbf{T}, C \mapsto \mathbf{F}\}$ is a satisfying assignment.
 - unsatisfiable: No, since there are satisfying assignments.
 - falsifiable: Yes, for example $\{A \mapsto \mathbf{T}, B \mapsto \mathbf{F}, C \mapsto \mathbf{F}\}$ does not satisfy φ .
 - valid: No, since some assignments do not satisfy φ .

(b)

$$\begin{aligned}\varphi &= ((A \vee \neg C) \rightarrow (\neg C \wedge (\neg A \vee B))) \\ &\equiv (\neg(A \vee \neg C) \vee (\neg C \wedge (\neg A \vee B))) && ((\rightarrow)\text{-elimination}) \\ &\equiv ((\neg A \wedge \neg \neg C) \vee (\neg C \wedge (\neg A \vee B))) && (\text{De Morgan}) \\ &\equiv ((\neg A \wedge C) \vee (\neg C \wedge (\neg A \vee B))) && (\text{double negation}) \\ &\equiv ((\neg A \vee (\neg C \wedge (\neg A \vee B))) \wedge (C \vee (\neg C \wedge (\neg A \vee B)))) && (\text{distributivity}) \\ &\equiv (((\neg C \wedge (\neg A \vee B)) \vee \neg A) \wedge ((\neg C \wedge (\neg A \vee B)) \vee C)) && (\text{commutativity}) \\ &\equiv ((\neg C \vee \neg A) \wedge ((\neg A \vee B) \vee \neg A)) \wedge ((\neg C \vee C) \wedge ((\neg A \vee B) \vee C)) && (\text{distributivity}) \\ &\equiv (\neg A \vee \neg C) \wedge (\neg A \vee B) \wedge (\neg A \vee B \vee C) && (\text{optional simplifying})\end{aligned}$$

Exercise 8.2 (0.5+1+2.5 marks)

- (a) Let φ and ψ be propositional logic formulas. If φ has 4 models and ψ has 3 models, what is the smallest and what is the largest number of models that $(\varphi \vee \psi)$ can have?
- (b) State a propositional logic formula over $\Sigma = \{A, B, C\}$ that expresses that exactly one variable in Σ is true.
- (c) Show with resolution that the CNF formula

$$(A \vee \neg B) \wedge (A \vee B \vee \neg C) \wedge (\neg A \vee \neg B \vee E) \wedge (B \vee C) \wedge (B \vee \neg C) \wedge (\neg B \vee D) \wedge \neg E$$

is unsatisfiable.

Note: This exercise was part of an exam question in 2024.

Solution:

- (a) between 4 and 7
- (b) $((A \vee B \vee C) \wedge \neg((A \wedge B) \vee (A \wedge C) \vee (B \wedge C)))$
- (c)
 - Resolve $(B \vee C)$ with $(B \vee \neg C)$ to get B .

- Resolve B with $(A \vee \neg B)$ to get A .
- Resolve $\neg E$ with $(\neg A \vee \neg B \vee E)$ to get $(\neg A \vee \neg B)$.
- Resolve $(\neg A \vee \neg B)$ with A to get $\neg B$.
- Resolve $\neg B$ with B to get $\perp$.

Exercise 8.3 (2+1 marks)

(a) Perform DPLL on

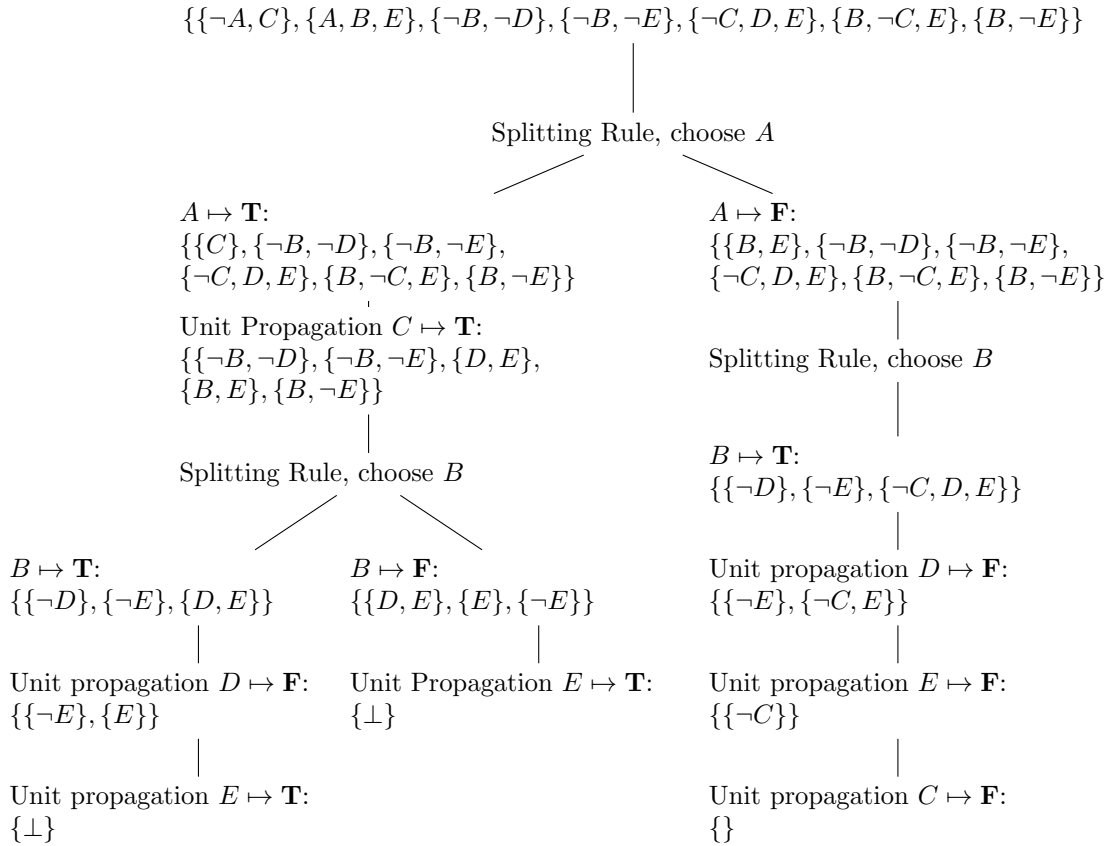
$$\{\{\neg A, C\}, \{A, B, E\}, \{\neg B, \neg D\}, \{\neg B, \neg E\}, \{\neg C, D, E\}, \{B, \neg C, E\}, \{B, \neg E\}\}$$

choosing variables alphabetically (also for unit propagation if there are several choices) and always assigning **T** first for splitting. Provide the application of all rules and the resulting tree as done in the lecture (chapter E4). What is the result of the algorithm?

(b) Would always assigning **F** first result in a smaller tree? Justify your answer.

Solution:

(a) DPLL produces the following tree:



The formula is satisfiable with interpretation $\{A \mapsto \mathbf{F}, B \mapsto \mathbf{T}, C \mapsto \mathbf{F}, D \mapsto \mathbf{F}, E \mapsto \mathbf{F}\}$.

(b) Yes. We would not need to explore the entire subtree rooted at “ $A \mapsto \mathbf{T}$ ”. We will need to explore a new subtree that assigns **F** to B immediately after assigning **F** to A , but this subtree will immediately be resolved through unit propagation since it will have clauses $\{E\}$ and $\{\neg E\}$. We then assign **T** to B (while still having assigned **F** to A), at which point the solution will be found through unit propagation.